Test Series Code

02

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | **DROPPER** | **M. MARKS: 720**

DATE: 02/11/2025

ANSWER KEY

1. (3)	46. (1)	91. (2)	136. (4)
2. (3)	47. (4)	92. (4)	137. (2)
3. (4)	48. (3)	93. (3)	138. (3)
4. (2)	49. (3)	94. (1)	139. (3)
5. (3)	50. (2)	95. (1)	140. (3)
6. (4)	51. (4)	96. (3)	141. (3)
7. (1)	52. (3)	97. (2)	142. (3)
8. (2)	53. (1)	98. (2)	143. (2)
9. (4)	54. (4)	99. (1)	144. (2)
10. (4)	55. (1)	100. (3)	145. (3)
11. (1)	56. (1)	101. (4)	146. (2)
12. (2)	57. (4)	102. (4)	147. (1)
13. (1)	58. (1)	103. (2)	148. (2)
14. (4)	59. (1)	104. (4)	149. (2)
15. (2)	60. (4)	105. (4)	150. (2)
16. (1)	61. (3)	106. (3)	151. (1)
17. (4)	62. (1)	107. (3)	152. (3)
18. (2)	63. (2)	108. (4)	153. (2)
19. (3)	64. (3)	109. (2)	154. (3)
20. (4)	65. (2)	110. (2)	155. (2)
21. (2)	66. (2)	111. (2)	156. (3)
22. (4)	67. (4)	112. (3)	157. (2)
23. (3)	68. (2)	113. (1)	158. (2)
24. (4)	69. (2)	114. (3)	159. (1)
25. (4)	70. (2)	115. (2)	160. (3)
26. (3)	71. (4)	116. (3)	161. (4)
27. (1)	72. (1)	117. (1)	162. (2)
28. (1)	73. (2)	118. (2)	163. (2)
29. (4)	74. (4)	119. (2)	164. (1)
30. (1)	75. (4)	120. (2)	165. (1)
31. (3)	76. (4)	121. (4)	166. (2)
32. (3)	77. (2)	122. (2)	167. (2)
33. (4)	78. (3)	123. (1)	168. (2)
34. (1)	79. (4)	124. (1)	169. (2)
35. (4)	80. (1)	125. (3)	170. (1)
36. (3)	81. (1)	126. (4)	171. (3)
37. (1)	82. (2)	127. (1)	172. (3)
38. (1)	83. (3)	128. (2)	173. (2)
39. (2)	84. (3)	129. (3)	174. (3)
40. (1)	85. (3)	130. (2)	175. (2)
41. (2)	86. (3)	131. (3)	176. (3)
42. (2)	87. (3)	132. (3)	177. (3)
43. (4)	88. (1)	133. (3)	178. (2)
44. (2)	89. (1)	134. (1)	179. (3)
45. (4)	90. (1)	135. (4)	180. (3)

1. (3)

For an ideal Atwood machine,

$a = \frac{(6-4)g}{6+4} = 2 \text{ m s}^{-2}$.

The blocks have equal and opposite accelerations, so $\vec{a}_2 = -\vec{a}_1 \Rightarrow |\vec{a}_1 - \vec{a}_2| = |2\vec{a}_1| = 2 \cdot 2 = 4 \text{ m s}^{-2}$.

(NCERT Ed. 2025-26, 11th, Page No. 70)

2. (3)

$y(t) = 5 + 5t \Rightarrow v = 5 \text{ m/s}, a = 0$

With zero acceleration, apparent weight $N = mg$. A spring balance reads mg as 50 kg-wt for a 50 kg man.

(NCERT Ed. 2025-26, 11th, Page No. 69)

3. (4)

Max possible acceleration is limited by friction: $a_{\max} = \mu g$. For minimum time, accelerate at $a_{\max}$ over distance $s : s = \frac{1}{2}at^2 \Rightarrow t = \sqrt{\frac{2s}{(\mu g)}} \propto \frac{1}{\sqrt{\mu}}$

(NCERT Ed. 2025-26, 11th, Page No. 62)

4. (2)

Atwood acceleration $a = \frac{(5-2)g}{5+2} = \frac{3g}{7}$.

CM acceleration $a_{\text{cm}} = \frac{5a + 2(-a)}{5+2}$

$= \frac{3a}{7} = \frac{9g}{49} = \frac{90}{49} \text{ m s}^{-2}$ (downward).

(NCERT Ed. 2025-26, 11th, Page No. 99)

5. (3)

Take right +. With wall speed $u_w = -4$ and ball $u = +6$, $v - u_w = -e(u - u_w)$

$\Rightarrow v + 4 = -\frac{1}{3}(6 - (-4)) = -\frac{10}{3}$

$\Rightarrow v = -\frac{22}{3}$ (left); speed $= \frac{22}{3} \text{ ms}^{-1}$

(NCERT Ed. 2025-26, 11th, Page No. 84)

6. (4)

$W = 2t^2 + 3t + 4$

$\frac{dW}{dt} = 4t + 3$

$(P)_{t=2} = 11W$

(NCERT Ed. 2025-26, 11th, Page No. 87)

7. (1)

Immediately after the impulse,

$v_{3\text{ kg}} = +10 \text{ m/s}, v_{2\text{ kg}} = 0$

$V_{\text{cm}} = \frac{3(10) + 2(0)}{3+2} = 6 \text{ m s}^{-1}$ to the right $(+\hat{i})$.

(NCERT Ed. 2025-26, 11th, Page No. 99)

8. (2)

$\vec{F} = -\nabla U$. With $U = -2x^2 y$

$F_x = -\frac{\partial U}{\partial x} = 4xy, \; F_y = -\frac{\partial U}{\partial y} = 2x^2$

So $\vec{F} = 4xy\hat{i} + 2x^2\hat{j}$

(NCERT Ed. 2025-26, 11th, Page No. 78)

9. (4)

1D elastic collision with

$m_1 = 8 \text{ kg}, m_2 = 2 \text{ kg}, u_2 = 0$

$v_1 = \frac{m_1 - m_2}{m_1 + m_2} u_1 = \frac{6}{10} u_1 = 0.6u_1.$

So $K_1' = \frac{1}{2} m_1 v_1^2 = 0.36 \frac{1}{2} m_1 u_1^2 = 0.36K_0$.

(NCERT Ed. 2025-26, 11th, Page No. 84)

10. (4)

By energy conservation from top to bottom:

$K_b = K_t + mg(2R) = \frac{1}{2} m(4^2) + 10 \cdot 2 = 8 + 20 = 28 \text{ J}$

$\Rightarrow v_b = \sqrt{56} \text{ m/s}$

At bottom: $T - mg = \frac{mv_b^2}{R}$

$\Rightarrow T = mg + mv_b^2 / R = 10 + 56 = 66 \text{ N}$

(NCERT Ed. 2025-26, 11th, Page No. 79)

11. (1)

Removing the top-right quarter (area = $\frac{1}{4}$ of the sheet) whose centroid is at $(4,2)\text{cm}$ from the origin shifts the COM by

$\vec{r}_{\text{rem}} = \frac{M\vec{0} - (M/4)(4,2)}{M - M/4}$

$= -\frac{1}{3}(4,2) = \left(-\frac{4}{3}, -\frac{2}{3}\right)\text{cm}$

(NCERT Ed. 2025-26, 11th, Page No. 98)

12. (2)

Static friction supplies centripetal force:

$m\omega^2 r \le \mu mg \Rightarrow r \le \frac{\mu g}{\omega^2} = \frac{0.5 \times 10}{5^2} = 0.2 \text{ m}.$

(NCERT Ed. 2025-26, 11th, Page No. 63)

[Contd...

13. (1)

Constant speed while descending

$\Rightarrow mg\sin\theta = \mu mg\cos\theta \Rightarrow \mu = \tan 37^\circ$.

While projected up, deceleration

$a = g(\sin\theta + \mu\cos\theta) = g(\sin\theta + \sin\theta)$

$= 2g\sin 37^\circ = 12 \text{ m s}^{-2}$

Stopping distance $s = \frac{u^2}{2a} = \frac{20^2}{2\times 12} = \frac{50}{3}$ m.

(NCERT Ed. 2025-26, 11th, Page No. 61)

14. (4)

Initially $T_2 = 1g = 10$ N and $T_1 = 2g + T_2 = 30$ N

Cut S_1 : on 2 kg , forces $= 20 + 10 = 30$ N

$a_1 = 30/2 = 15 \text{ m s}^{-2}$.

Cut S_2 : on 2 kg , forces $= 30 - 20 = 10$ N

$\Rightarrow a_2 = 10/2 = 5 \text{ m s}^{-2}$.

$|a_1| : |a_2| = 15 : 5 = 3 : 1$.

(NCERT Ed. 2025-26, 11th, Page No. 59)

15. (2)

$T - mg = ma$

$\Rightarrow T = m(g + a) = 25{,}000(10 + 6) = 4.0 \times 10^5$ N

(NCERT Ed. 2025-26, 11th, Page No. 69)

16. (1)

$s = \frac{1}{3}t^3 \Rightarrow v = t^2$.

At $t = 0, v = 0$; at $t = 2, v = 4$ m/s.

Work $W = \Delta K = \frac{1}{2}m(v^2 - v_0^2) = \frac{1}{2}\cdot 3 \cdot 16 = 24$ J.

(NCERT Ed. 2025-26, 11th, Page No. 76)

17. (4)

Area under *F*-*x* graph = Work done

work done = $(5 \times 4) + \left(\frac{1}{2}\times 3.14 \times 4\right) = 26.28$ Joule

(NCERT Ed. 2025-26, 11th, Page No. 74)

18. (2)

With constant force *F*, $a = \frac{F}{m} = \text{const} \Rightarrow v = at \propto t$

Power $P = Fv \propto t$.

(NCERT Ed. 2025-26, 11th, Page No. 83)

19. (3)

Wind power $P = \frac{1}{2}\rho A v^3$.

With $v = 36 \text{ km h}^{-1} = 10 \text{ ms}^{-1}$

$P_{\text{wind}} = 0.5(1.2)(30)(10^3) = 18$ kW

25% conversion $\Rightarrow 0.25 \times 18 = 4.5$ kW

(NCERT Ed. 2025-26, 11th, Page No. 91)

20. (4)

$F_c = m\omega^2 r$. With $m = 1$ kg, $r = 2$ m, $f = 1$rps

$\Rightarrow \omega = 2\pi \Rightarrow F = 1\cdot(2\pi)^2 \cdot 2 = 8\pi^2 \text{N} \neq 4\pi^2 \text{N}$.

So I is incorrect, II is correct.

(NCERT Ed. 2025-26, 11th, Page No. 63)

21. (2)

From $0 - 4$ s, slope $= \frac{8}{4} = 2$ m/s $\Rightarrow v = 2$ m/s.

For $t > 4, x$ is constant $\therefore v = 0$.

Impulse $J = \Delta p$

$= 2(0 - 2) = -4$ Ns (opposite to motion).

(NCERT Ed. 2025-26, 11th, Page No. 56)

22. (4)

$mg = 10$ N, $mg\sin 30^\circ = 5$ N,

$N = mg\cos 30^\circ = 5\sqrt{3}$ N

Actual friction $= 5$ N (up the plane)

Limiting friction $= \mu N = 0.8 \times 5\sqrt{3} = 4\sqrt{3}$ N

Contact force $= \sqrt{N^2 + F^2} = \sqrt{(5\sqrt{3})^2 + 5^2} = 10$ N

For just moving down: $P + 5 = 4\sqrt{3} \Rightarrow P \approx 1.93$ N

(NCERT Ed. 2025-26, 11th, Page No. 61)

23. (3)

For an elastic collision of identical masses with one at rest, the two final velocities are perpendicular,

$v_1\sin 37 = v_2\sin 53$

$3v_1 = 4v_2$

Conservation of momentum in *x* direction,

$5 = \frac{16v_2}{15} + \frac{3v_2}{5} \Rightarrow v_2 = 3$ m/s

(NCERT Ed. 2025-26, 11th, Page No. 85)

24. (4)

Viscous (drag) force is velocity-dependent and dissipative; work own a closed path isn't zero $\Rightarrow$ nonconservative.

(NCERT Ed. 2025-26, 11th, Page No. 82)

[Contd...

25. (4)

Net work done is zero (starts and ends at rest).

Gravity work $= mgL\sin\theta$. Friction acts only on lower $L/3$: loss $= \mu mg\cos\theta(L/3)$.

Set equal $\Rightarrow mgL\sin\theta = \mu mg\cos\theta\left(\frac{L}{3}\right)$

$\Rightarrow \mu = 3\tan\theta$

(NCERT Ed. 2025-26, 11th, Page No. 82)

26. (3)

Equilibrium occur where $F = 0$ at x_1 and x_3 (not at x_2).

Stability from slope: stable if $\frac{dF}{dx} < 0$

At x_1: slope negative $\Rightarrow$ stable. At x_3: slope positive $\Rightarrow$ unstable.

(NCERT Ed. 2025-26, 11th, Page No. 88)

27. (1)

Net external force $= F_1 - F_2 \Rightarrow$ rod's acceleration $a = \frac{F_1 - F_2}{M}$ (to the right).

For the left segment of length y (mass $= \frac{M}{L}y$), force balance along the rod gives

$T(y) - F_2 = \left(\frac{M}{L}y\right)a$

$T(y) = F_2 + \frac{M}{L}y\cdot\left(\frac{F_1 - F_2}{M}\right)$

$= F_2 + \frac{F_1 - F_2}{L}y = F_1\frac{y}{L} + F_2\left(1 - \frac{y}{L}\right)$

(NCERT Ed. 2025-26, 11th, Page No. 54)

28. (1)

$\Delta\vec{p} = m(\vec{v} - \vec{u}) = 1\left[\left(-\left(4\hat{i} + 3\hat{j}\right)\right) - \left(4\hat{i} + 3\hat{j}\right)\right]$

$= \left(-8\hat{i} - 6\hat{j}\right)$

$|\Delta\vec{p}| = \sqrt{8^2 + 6^2} = 10$ N/s

(NCERT Ed. 2025-26, 11th, Page No. 53)

29. (4)

Hanging part has length $\frac{l}{3} \Rightarrow$ mass $\frac{m}{3}$ and its COM is at $\frac{l}{6}$ below the table edge. Lifting it onto the table raises it by $\frac{l}{6}$:

$\Delta U = \left(\frac{m}{3}\right)g\left(\frac{l}{6}\right) = \frac{mgl}{18}$.

(NCERT Ed. 2025-26, 11th, Page No. 77)

30. (1)

Explosion is internal $\Rightarrow$ CM follows the same projectile path as if no explosion.

$\frac{1}{2}m\times(20)^2 + 0 = \frac{1}{2}mv_2^2 + mg(2)$

$\frac{400}{2} = \frac{(v_2)^2}{2} + 20$

$v_2 = \sqrt{360}$ m/s

(NCERT Ed. 2025-26, 11th, Page No. 100)

31. (3)

$P = \frac{\text{work}}{\text{time}}$, and $[W] = [FL]$

$[W] = \left[MLT^{-2}\right]\cdot[L] = \left[ML^2T^{-2}\right]$

Hence $[P] = \left[ML^2T^{-3}\right]$.

(NCERT Ed. 2025-26, 11th, Page No. 83)

32. (3)

For a bounce on a fixed surface, rebound height $h' = e^2h$.

Given $e = 0.5$ and $h' = 1$ m $\Rightarrow h = \frac{1}{(0.5)^2} = 4$ m.

(NCERT Ed. 2025-26, 11th, Page No. 84)

33. (4)

Work on a body = force × displacement of that body's point of application.

The road doesn't move, so despite the 200 N friction, the road's displacement is zero $\Rightarrow$ work done by the cycle on the road is 0.

(NCERT Ed. 2025-26, 11th, Page No. 74)

34. (1)

Classically allowed where $K = E - U(x) \geq 0$

$\Rightarrow E \geq U(x)$.

Here $U(x) > E$ for $x < a$ and $x > b$, so the particle cannot be found there.

(NCERT Ed. 2025-26, 11th, Page No. 88)

35. (4)

For a banked curve with friction μ the no-slip speed range is $v_{\min} = \sqrt{\frac{Rg(\tan\theta - \mu)}{1 + \mu\tan\theta}}$

(NCERT Ed. 2025-26, 11th, Page No. 64)

[Contd...

36. (3)

Max static frictions: $f_{A,max} = 0.5(10g) = 50$ N,

$f_{B,max} = 0.5(5g) = 25$ N.

For no slip on A: $65 - (T + f_A) = 0$

At the threshold $f_A = 50$, So $T = 15$ N

For block B in equilibrium: $T = f_B = 15$ N.

(NCERT Ed. 2025-26, 11th, Page No. 61)

37. (1)

For the block to stay at rest on the smooth plane, component along the plane must vanish: $F\cos\theta = mg\sin\theta \Rightarrow F = mg\tan\theta = 20/\sqrt{3}$ N.

The normal then balances the perpendicular components

$N = mg\cos\theta + F\sin\theta$.

Hence all components cancel $\Rightarrow$ net force on the block is zero.

(NCERT Ed. 2025-26, 11th, Page No. 54)

38. (1)

Ground is smooth; if blocks move together, $a = \frac{F}{2m} = \frac{1.5\mu mg}{2m} = \frac{3}{4}\mu g$.

Friction needed to accelerate B is $f = ma = \frac{3}{4}\mu mg < \mu mg = f_{max}$, so no slipping occurs $\Rightarrow$ both blocks share $a = \frac{3}{4}\mu g$.

(NCERT Ed. 2025-26, 11th, Page No. 60)

39. (2)

From the graph, $F = 4$ N for $0 \le x \le 2$ m.

Work $W = \int_0^2 F dx = 4 \times 2 = 8$ J.

(NCERT Ed. 2025-26, 11th, Page No. 76)

40. (1)

For a perfectly inelastic hit, final speed $V = \frac{2m \cdot v + m(-v)}{3m} = \frac{v}{3}$.

Initial KE $= \frac{3}{2}mv^2$

Final KE $= \frac{1}{2}(3m)\left(\frac{v}{3}\right)^2 = \frac{1}{6}mv^2$.

Loss $= \left(\frac{3}{2} - \frac{1}{6}\right)mv^2 = \frac{4}{3}mv^2$.

(NCERT Ed. 2025-26, 11th, Page No. 84)

41. (2)

$K = \frac{p^2}{2m}$. With equal momenta, $K \propto \frac{1}{m}$

$\Rightarrow \frac{K_m}{K_M} = \frac{M}{m}$ (lighter mass has larger KE).

(NCERT Ed. 2025-26, 11th, Page No. 74)

42. (2)

At constant speed. motor force must balance weight + friction:

$F = mg + f = (1800 \times 10) + 2000 = 20000$ N

Power $P = Fv = 20000 \times 1 = 2.0 \times 10^4$ W $= 20$ kW.

(NCERT Ed. 2025-26, 11th, Page No. 83)

43. (4)

In an isolated system (no external impulse), internal forces are equal and opposite $\rightarrow$ total linear momentum is conserved for elastic, inelastic, or perfectly inelastic collisions.

(NCERT Ed. 2025-26, 11th, Page No. 84)

44. (2)

Only external force is gravity: $F_{ext} = (2m + 4m)g$ downward.

So $a_{CM} = \frac{F_{ext}}{(2m + 4m)} = g = 10$ m s^{-2} downward

(independent of their speeds).

(NCERT Ed. 2025-26, 11th, Page No. 99)

45. (4)

$W_1 = \frac{1}{2}kx^2 = W \Rightarrow k = \frac{2W}{x^2}$

Extra work from x to $2x$:

$\Delta W = \frac{1}{2}k\left[(2x)^2 - x^2\right] = \frac{1}{2}k(3x^2) = 3W$.

(NCERT Ed. 2025-26, 11th, Page No. 81)

[Contd...

46. (1)

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\frac{1}{\lambda} = R \times (1)^2 \times \left(\frac{1}{1^2} - \frac{1}{\infty^2}\right)$$

$$\lambda = \frac{1}{R}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 45)

47. (4)

	$AB \rightarrow$	$A^+ +$	B^-
(i)	1	0	0
(f)	–	0.75	0.75

	$3AB \rightarrow$	$(AB)_3$
(i)	1	0
(f)	–	$\frac{0.25}{3}$

van't Hoff factor $(i) = 0.75 + 0.75 + \frac{0.25}{3}$

$= 1.58$

(NCERT Ed. 2025-26 12th Part-I Page No. 24)

48. (3)

$$\frac{\nu_1}{\nu_2} = \frac{\lambda_2}{\lambda_1} = \frac{2.05 \times 10^{-5}}{4.1 \times 10^{-5}} = \frac{1}{2}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 41)

49. (3)

$$\Delta T_f = iK_f\, m$$

$$1 = 2 \times 1.86 \times \frac{w}{58.5} \times \frac{1000}{1860}$$

$$w = 29.25\, g$$

(NCERT Ed. 2025-26 12th Part-I Page No. 24)

50. (2)

$$E_{photon} = W_0 + KE$$

$$\frac{hc}{\lambda} = W_0 + 1\, eV$$

$$\frac{2hc}{\lambda} = W_0 + 4\, eV$$

$$\therefore\ 2(W_0 + 1\, eV) = W_0 + 4\, eV$$

$$W_0 = 2\ eV$$

(NCERT Ed. 2025-26 11th Part-I Page No. 42)

51. (4)

- Osmosis always take place from hypotonic to hypertonic solutions.
- The release of dissolved nitrogen gas in the blood creates a medical condition known as bends.

(NCERT Ed. 2025-26 12th Part-I Page No. 23, 08)

52. (3)

Half filled and completely filled subshells have extra stability. It is due to relatively small shielding, smaller coulombic repulsion energy and larger exchange energy.

(NCERT Ed. 2025-26 11th Part-I Page No. 65)

53. (1)

$$\Delta T_b = K_b m$$

$$0.26 = 0.52\ m$$

$$m = \frac{1}{2}\, mol\, kg^{-1}$$

Now, $\frac{p^0 - p_s}{p^0} = \frac{n}{n+N} \simeq \frac{n}{N}$

$$\frac{760 - p_s}{760} = \frac{1}{\left(\frac{2000}{18}\right)}$$

$$p_s = 753.16\, mm\, of\, Hg$$

(NCERT Ed. 2025-26 12th Part-I Page No. 16)

54. (4)

$Be(Z = 4) : 1s^2 2s^2$

last electron of Be atom is present in 2s orbital; so for that electron

$$n = 2,\ \ell = 0,\ m = 0,\ s = \pm\frac{1}{2}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 63)

55. (1)

- Higher the K_H, lesser will be the solubility gas.
- More polar gases are more soluble.
- Inert gases have very low solubility in water.

(NCERT Ed. 2025-26 12th Part-I Page No. 08)

56. (1)

ν: Radio waves < UV-rays < X-rays

(NCERT Ed. 2025-26 11th Part-I Page No. 38)

[Contd...

57. (4)

Amalgam of mercury with sodium is a type of liquid in solid solution.

(NCERT Ed. 2025-26 12th Part-I Page No. 02)

58. (1)

$$\text{Energy of each photon} = \frac{hc}{\lambda}$$

$$\text{Kinetic energy of each atom} = \frac{\frac{hc}{\lambda} - \frac{60.22\times10^3}{N_A}}{2}$$

$$= \frac{\left(\frac{6.62\times10^{-34}\times3\times10^8}{6620\times10^{-10}} - \frac{60.22\times10^3}{6.022\times10^{23}}\right)}{2} = 10^{-19}\,\text{J}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 41)

59. (1)

$$\text{Molality(m)} = \frac{\left(\frac{20}{180}\right)}{80\times10^{-3}}$$

$$= 1.39\,\text{mol kg}^{-1}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 05)

60. (4)

The energy of an electron in one electron species is determined solely by principal quantum number (n). As the n increases, energy of orbitals increases.

(NCERT Ed. 2025-26 11th Part-I Page No. 60)

61. (3)

$$M_1 = \frac{\%\text{ by mass of solute} \times \text{d(g/mL)} \times 10}{\text{Molar mass of slute}}$$

$$M_1 = \frac{9.8\times2\times10}{98} = 2\,\text{M}$$

Now, $M_1V_1 = M_2V_2$

$2\times V_1 = 0.1\times200$

$V_1 = 10\,\text{mL}$

(NCERT Ed. 2025-26 12th Part-I Page No. 04)

62. (1)

$$r_n = \frac{52.9\,n^2}{Z}\,\text{pm}$$

$$r_2 = \frac{52.9\times3^2}{4}\,\text{pm} = 119.03\,\text{pm}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 48)

63. (2)

$[H^+] = C\alpha$

$10^{-1} = 0.1\,\alpha$

$\alpha = 1$

$$\pi = iCRT = 2\times0.1\times R\times T = 0.2\,RT$$

(NCERT Ed. 2025-26 12th Part-I Page No. 24)

64. (3)

Interference phenomenon can explain wave nature.

(NCERT Ed. 2025-26 11th Part-I Page No. 39)

65. (2)

$$p_{total} = p_A^0X_A + p_B^0(1-X_A) = p_B^0 + X_A(p_A^0 - p_B^0)$$

and $Y_A = \frac{p_A}{p_{total}} = \frac{p_A^0X_A}{p_{total}}$

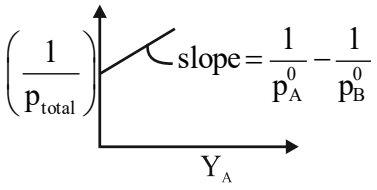
$$\Rightarrow X_A = \frac{Y_Ap_{total}}{p_A^0}$$

$$\therefore\ p_{total} = p_B^0 + \frac{Y_Ap_{total}}{p_A^0}(p_A^0 - p_B^0)$$

$$1 = \frac{p_B^0}{p_{total}} + \frac{Y_A}{p_A^0}(p_A^0 - p_B^0)$$

$$\frac{p_B^0}{p_{total}} = 1 - \frac{Y_A}{p_A^0}(p_A^0 - p_B^0)$$

$$\frac{1}{p_{total}} = \frac{1}{p_B^0} + Y_A\left(\frac{1}{p_A^0} - \frac{1}{p_B^0}\right)$$



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66. (2)

Number of neutrons present in the nucleus have very little effect on the chemical properties of an element.

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67. (4)

For ideal solutions,

$\Delta_{mix}G < 0$ and $\Delta_{mix}S > 0$

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68. (2)

For visible region, $n_2 = 5$ and $n_1 = 2$

$\therefore$ so 3 lines are obtain

$5 \rightarrow 2$

$4 \rightarrow 2$

$3 \rightarrow 2$

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69. (2)

- With increase in temperature, vapour pressure increases.
- Vapour pressure will be same for non-volatile, non-electrolyte aqueous isotonic solutions.

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70. (2)

$Cr(Z=24):[Ar]3d^5 4s^1:$ Unpaired $e^- = 6$

$Pd(Z=46):[Kr]4d^{10}:$ Unpaired $e^- = 0$

$Ni(Z=28):[Ar]3d^8 4s^2:$ Unpaired $e^- = 2$

$Ag(Z=47):[Kr]4d^{10} 5s^1:$ Unpaired $e^- = 1$

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71. (4)

$_1^1H: p = 1$ and $n = 1 - 1 = 0$

$_1^2D: p = 1$ and $n = 2 - 1 = 1$

$_1^3T: p = 1$ and $n = 3 - 1 = 2$

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72. (1)

Molality, Mole fraction and % by (mass/mass) terms do not depend on temperature as they not involve volume term.

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73. (2)

$$\lambda = \frac{h}{\sqrt{2mKE}}$$

$$\lambda = \frac{6.62 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 9.1 \times 10^{-25}}}$$

$$\lambda = 0.51 \times 10^{-6}\ m$$

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74. (4)

$\Delta T_b = K_b$ molality

if, molality (m) = 1,

$\Delta T_b = K_b$

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75. (4)

Electron spin quantum number is not obtain by solving Schrodinger wave equation.

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76. (4)

$$\frac{p^0 - p_x}{p^0} = X_{solute}$$

Case-I: $\frac{10}{p^0} = 0.2$

$$\Rightarrow p^0 = \frac{10}{0.2}$$

$= 50$ mm of Hg

Case-II: $\frac{5}{50} = X_{solute}$

$$\therefore\ X_{solvent} = 1 - \frac{5}{50} = 0.90$$

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77. (2)

$$r = 0.529 \times \frac{n^2}{Z}\ \text{Å}$$

$$2.116 = 0.529 \times \frac{n^2}{1}$$

$n = 2$

angular momentum

$$= \frac{nh}{2\pi}$$

$$= \frac{2h}{2\pi} = \frac{h}{\pi}$$

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78. (3)

Aquatic species are more comfortable in cold water rather than in warm water because on increasing temperature solubility of oxygen in water decreases.

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[Contd...

79. (4)

- According to Aufbau principle, 2s orbital is filled before 2p orbitals.
- According to Hund's rule, in p-subshell electron pairing take place only when each orbital of subshell attain one electron each in parallel spin.

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80. (1)

10^6 g of solution contain 10 g $CHCl_3$.

$$\therefore \text{ \% by mass} = \frac{10}{10^6} \times 100 = 1 \times 10^{-3}$$

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81. (1)

$$KE = +13.6 \frac{Z^2}{n^2} \text{ eV}$$

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82. (2)

Mass % of resulting solution

$$= \frac{\left[100 \times \frac{50}{100} + 200 \times \frac{60}{100}\right]}{100 + 200} \times 100$$

$$= 56.66\%$$

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83. (3)

It is not possible to draw a boundary surface diagram of rigid size in which the probability of finding the electron is 100% because the probability density $|\psi|^2$ has always some value, howsoever small it may be, at infinite distance from the nucleus.

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84. (3)

Reverse osmosis is used in desalination of sea water.

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85. (3)

- radial nodes $= n - \ell - 1 = 4 - 0 - 1 = 3$
- angular nodes $= \ell = 0$

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86. (3)

$$\Delta T_f = i \times K_f \times m$$

$$0.372 = i \times 1.86 \times 0.1$$

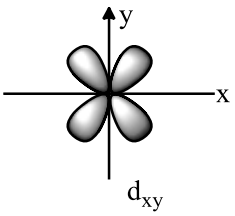
$$i = 2$$

$$[Cr(NH_3)_4Cl_2]Cl \rightarrow [Cr(NH_3)_4Cl_2]^+ + Cl^-$$

$$i = 2$$

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87. (3)



In d_{xy} orbital electron density is along the axis

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88. (1)

$$C_1 = C_2$$

$$\frac{\left(\frac{1}{60}\right)}{100 \times 10^{-3}} = \frac{\left(\frac{2}{M}\right)}{100 \times 10^{-3}}$$

$$M = 120 \text{ g mol}^{-1}$$

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89. (1)

$$\Delta v = \frac{h}{4\pi \Delta x m}$$

$$= \frac{6.62 \times 10^{-34}}{4 \times 3.14 \times 1 \times 10^{-10} \times 9.1 \times 10^{-31}}$$

$$= 5.8 \times 10^5 \text{ ms}^{-1}$$

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90. (1)

Binary mixture of ethanol and water shows positive deviation from Raoult's law. Vapour pressure of solution is more than vapour pressure of pure ethanol and water, so the boiling point of solution is less than boiling point of pure ethanol and water. Hence it form minimum boiling azeotrope.

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[Contd...

91. (2)

S or synthesis phase marks the period during which DNA synthesis or replication takes place. During this time the amount of DNA per cell doubles. If the initial amount of DNA is denoted as 2C then it increases to 4C. However, there is no increase in the chromosome number; if the cell had diploid or 2n number of chromosomes at G_1, even after S phase the number of chromosomes remains the same, i.e., 2n. In animal cells, during the S phase, DNA replication begins in the nucleus, and the centriole duplicates in the cytoplasm. S phase is present in both mitotic and meiotic divisions.

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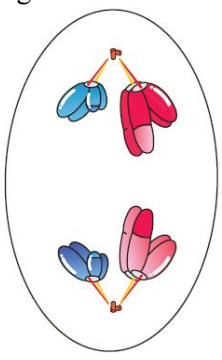
92. (4)

Although cell growth (in terms of cytoplasmic increase) is a continuous process, DNA synthesis occurs only during one specific stage in the cell cycle.

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93. (3)

The stage shown in the diagram is anaphase I. The homologous chromosomes separate, while sister chromatids remain associated at their centromeres. The homologous chromosomes have already undergone crossing over in prophase I. Bivalents (paired homologous chromosomes) are formed in prophase I and align at the equator in metaphase I. In the stage shown (anaphase I), they have already separated. Equational division refers to Mitosis (or meiosis II), where sister chromatids separate. The diagram shows a "reduction division" (meiosis I) where homologous chromosomes separate.



Anaphase I

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94. (1)

Based on the sub-stages of Prophase I of Meiosis, let's break down the sequence of the given events:

E. Compaction of chromosomes starts so that they can be gradually visualised under the light microscope. This is the first event and characterizes the Leptotene stage.

C. The chromosomes that have similar gene loci come together and form pairs. This process is called synapsis, which is the defining event of the Zygotene stage.

B. The enzyme recombinase is actively catalysing the process of crossing over. Crossing over between non-sister chromatids of homologous chromosomes, mediated by the enzyme recombinase, occurs during the Pachytene stage.

A. The homologous chromosomes remain attached only at the sites of crossing over. After crossing over is complete, the synaptonemal complex dissolves, and the homologous chromosomes begin to separate but remain attached at X-shaped structures called chiasmata (the sites of crossing over). This is characteristic of the Diplotene stage.

D. Chromatin is completely condensed into chromosomes. The final condensation of chromosomes and terminalisation of chiasmata occur in the last stage, Diakinesis.

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95. (1)

Some cells in the adult animals do not appear to exhibit division (e.g., heart cells) and many other cells divide only occasionally, as needed to replace cells that have been lost because of injury or cell death. These cells that do not divide further exit G_1 phase to enter an inactive stage called quiescent stage (G_0) of the cell cycle. Cells in this stage remain metabolically active but no longer proliferate unless called on to do so depending on the requirement of the organism.

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96. (3)

Meiosis ensures the production of haploid phase in the life cycle of sexually reproducing organisms whereas fertilisation restores the diploid phase.

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97. (2)

A male honey bee produces gametes (sperm) through mitosis, not meiosis. This is because it is already haploid and needs to produce haploid gametes.

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98. (2)

In the G_1 phase the cell has 32 chromosomes. At this stage, each chromosome consists of a single, unreplicated DNA molecule (a single chromatid). During the S Phase: The cell undergoes DNA

[Contd...

replication. Each of the 32 chromosomes is duplicated. After S phase, a replicated chromosome is still counted as one chromosome as long as its two identical sister chromatids are joined at a single centromere. However, each of these 32 chromosomes is now composed of a pair of identical sister chromatids. Therefore, the number of pairs of sister chromatids is equal to the number of chromosomes present, which is 32.

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99. (1)

Telophase, in both mitosis and meiosis I, is characterized by the reformation or reappearance of the nuclear envelope and nucleolus. The disintegration of these structures occurs during prophase. In Meiotic Anaphase I, homologous chromosomes separate. In Mitotic Anaphase, sister chromatids separate. In Meiotic Metaphase I, the number of chromosomes is double than that in metaphase II.

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100. (3)

(D) Cell cycle in a human cell in culture: 24 hours = 1440 minutes

(B) Interphase in a human cell in culture: 23 hours = 1380 minutes

(A) Cell cycle in yeast: 90 minutes

(E) Interphase in yeast: This will be the vast majority of the 90-minute yeast cell cycle. While the exact time isn't given, it will be slightly less than 90 minutes (e.g., ~80-85 minutes).

(F) M phase in a human cell in culture: 1 hour = 60 minutes

(C) M phase in yeast: This will be a very small fraction of the 90-minute cycle (e.g., ~5-10 minutes).

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101. (4)

Since the number of chromosomes in the parent and progeny cells is the same, mitosis is also called as equational division. Therefore, by undergoing mitosis, a cell having 16 chromosomes will produce two daughter cells having 16 chromosomes each.

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102. (4)

The cell cycle is divided into two basic phases: Interphase and M Phase (Mitosis phase). The interphase, though called the resting phase, is the time during which the cell is preparing for division by undergoing both cell growth and DNA replication in an orderly manner. The interphase is divided into three further phases: G_1 phase (Gap 1), S phase (Synthesis) and G_2 phase (Gap 2). M phase is the most dramatic period of the cell cycle. Though for convenience mitosis has been divided into four stages of nuclear division (karyokinesis), it is very essential to understand that cell division is a progressive process and very clear-cut lines cannot be drawn between various stages. Prophase of the first meiotic division is typically longer and more complex when compared to prophase of mitosis. It has been further subdivided into the following five phases based on chromosomal behaviour, i.e., Leptotene, Zygotene, Pachytene, Diplotene and Diakinesis.

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103. (2)

In some organisms karyokinesis (nuclear division) is not followed by cytokinesis (cytoplasmic division) as a result of which multinucleate condition arises leading to the formation of syncytium (e.g., liquid endosperm in coconut).

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104. (4)

D. The nuclear membrane was intact and a non-membranous cytoplasmic organelle was duplicating. Phase: Interphase. This is the preparatory phase before any division begins. The cell grows, DNA is replicated, and organelles are duplicated. The intact nuclear membrane is a key indicator of a cell that is not yet in the process of active division (M-phase).

B. Out of four chromatids, two sets of two chromatids each were formed: Anaphase I of Meiosis. This statement describes the separation of homologous chromosomes. A pair of homologous chromosomes consists of four chromatids. During Anaphase I, these homologous chromosomes (each still made of two sister chromatids) are pulled apart, resulting in "two sets of two chromatids" moving to opposite poles.

E. Two cells were seen without any major event of karyokinesis or cytokinesis: End of Meiosis I / Telophase I & Cytokinesis I / Interkinesis. Meiosis I concludes with the formation of two distinct haploid cells. This observation captures the state immediately following the first meiotic division. The two cells exist before entering the next phase of division.

A. He was able to study the size and shape of the chromosomes arranged on a single equatorial plate: Metaphase II of Meiosis. This is best observed during Metaphase II. In this stage, the chromosomes in each of the two daughter cells align individually at the metaphase plate. They are highly condensed and clearly visible, making it an ideal time to study their morphology (size and shape).

C. A cluster of four cells was formed: End of Meiosis II / Telophase II & Cytokinesis II. This is the final product of the entire meiotic process. The two cells produced from Meiosis I each divide, resulting in a total of four haploid cells, often found in a tetrad. This signifies the completion of cell division.

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105. (4)

Meiosis involves two sequential cycles of nuclear and cell division called meiosis I and meiosis II but only a single cycle of DNA replication. Meiosis I is initiated after the parental chromosomes have replicated to produce identical sister chromatids at the S phase.

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106. (3)

The key finding is a single large cell containing two genetically identical, diploid nuclei. This indicates that karyokinesis was successful i.e. the cell successfully completed all the stages of mitosis (prophase, metaphase, anaphase, and telophase), resulting in the formation of two separate and complete nuclei, but cytokinesis failed i.e. after the nucleus divided, the cytoplasm did not divide to form two separate daughter cells. This resulted in a single cell that is now binucleate. In plant cells, cytokinesis is achieved by the formation of a cell plate. If a drug inhibits the formation or function of this cell plate precursor, karyokinesis will finish, but the cell will be unable to physically divide. This perfectly explains the formation of a single, large cell with two nuclei.

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107. (3)

The interphase, though called the resting phase, is the time during which the cell is preparing for division by undergoing both cell growth and DNA replication in an orderly manner.

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108. (4)

Interkinesis: This is the short-lived stage that occurs between meiosis I and meiosis II.

G_1 Phase (Gap 1): This phase is the interval between the completion of mitosis and the beginning of DNA replication (S phase).

Diplotene: The stage which may last for months or years in oocytes of some vertebrates.

Diakinesis: Stage of meiosis I when spindle apparatus assembles.

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109. (2)

Starts of S phase: The cell has 4n chromosomes and 4C DNA.

First stage of meiosis I: Prophase I: This stage occurs after DNA replication (S-phase) but before the first meiotic division. The cell will have 4n chromosomes and 8C DNA.

Stage between meiosis I and meiosis II: Interkinesis: This is the brief stage between Meiosis I and Meiosis II. It represents the state of the cell immediately after the first meiotic division. Each cell will have 2n chromosomes and 4C DNA.

After cytokinesis of meiosis II: This is the final state of the four resulting gametes or spores after both meiotic divisions are complete. Each cell will have 2n chromosomes and 2C DNA.

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110. (2)

The sequence of events by which a cell duplicates its genome, synthesises the other constituents of the cell and eventually divides into two daughter cells is termed cell cycle. According to the cell theory, cells arise from pre-existing cells. All the processes, i.e., cell division, DNA replication, and cell growth have to take place in a coordinated way to ensure correct division and formation of progeny cells containing intact genomes. Cell cycle ends with cytokinesis. Any sexually reproducing organism (small as well as large) starts its life cycle from a single-celled zygote.

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111. (2)

M phase is the most dramatic period of the cell cycle, involving a major reorganisation of virtually all components of the cell. Since the number of chromosomes in the parent and progeny cells is the same, it is also called as equational division. Though for convenience mitosis has been divided into four stages of nuclear division (karyokinesis), it is very essential to understand that cell division is a

progressive process and very clear-cut lines cannot be drawn between various stages. The M phase begins with Prophase, which is characterized by the condensation of chromatin into visible, compact chromosomes. The M phase ends with Telophase (followed by cytokinesis), where the separated chromosomes arrive at the poles and begin to decondense back into chromatin.

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112. (3)

Anaphase II: It begins with the simultaneous splitting of the centromere of each chromosome (which was holding the sister chromatids together), allowing them to move toward opposite poles of the cell. This is similar to mitotic anaphase.

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113. (1)

Mitosis or the equational division is usually restricted to the diploid cells only. However, in some lower plants and in some social insects haploid cells also divide by mitosis. Mitosis usually results in the production of diploid daughter cells with identical genetic complement. The growth of multicellular organisms is due to mitosis. Cell growth results in disturbing the ratio between the nucleus and the cytoplasm. It therefore becomes essential for the cell to divide to restore the nucleo-cytoplasmic ratio. A very significant contribution of mitosis is cell repair. The cells of the upper layer of the epidermis, cells of the lining of the gut, and blood cells are being constantly replaced. Mitotic divisions in the meristematic tissues – the apical and the lateral cambium, result in a continuous growth of plants throughout their life.

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114. (3)

Some cells in the adult animals do not appear to exhibit division (e.g., heart cells) and many other cells divide only occasionally, as needed to replace cells that have been lost because of injury or cell death. These cells that do not divide further exit G_1 phase to enter an inactive stage called quiescent stage (G_0) of the cell cycle. In the G_1 phase the cell is actively synthesising proteins required for DNA replication.

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115. (2)

The complex formed by a pair of synapsed homologous chromosomes is called a bivalent or a tetrad (zygotene). Pachytene is characterised by the appearance of recombination nodules. The final stage of meiotic prophase I is diakinesis. This is marked by terminalisation of chiasmata. Anaphase II: It begins with the simultaneous splitting of the centromere of each chromosome.

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116. (3)

Kinetochores are disc-shaped structures located on the centromeres of chromosomes. They are the attachment sites for the spindle fibers (microtubules), not for the centrioles themselves.

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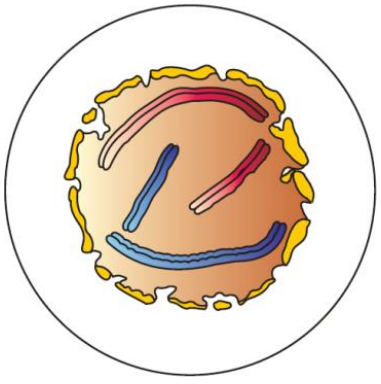
117. (1)

In an animal cell, cytokinesis is achieved by the appearance of a furrow in the plasma membrane. In plant cells, wall formation starts in the centre of the cell and grows outward to meet the existing lateral walls. The formation of the new cell wall begins with the formation of a simple precursor, called the cell-plate that represents the middle lamella between the walls of two adjacent cells. At the time of cytoplasmic division, organelles like mitochondria and plastids get distributed between the two daughter cells. The order of events in the M phase is nuclear division (karyokinesis) first, followed by cytoplasmic division (cytokinesis).

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118. (2)

The diagram represents the first stage of karyokinesis in an animal cell. The complete disintegration of the nuclear envelope marks the start of the second phase of mitosis i.e. metaphase, hence the chromosomes are spread through the cytoplasm of the cell. By this stage, condensation of chromosomes is completed and they can be observed clearly under the microscope. The chromosomes reach an extremely extended state at the end of telophase.



Late prophase

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119. (2)

The production of offspring by sexual reproduction includes the fusion of two gametes, each with a complete haploid set of chromosomes. Gametes are formed from specialised diploid cells. A zygote is, by definition, the cell that results from the fusion of two gametes. Since gametes are always haploid (n), the fusion of a male gamete (n) and a female gamete (n) will always result in a diploid (2n) zygote.

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120. (2)

He is observing the male germ cells. This is incorrect because spermatogenesis produces equal-sized cells.

He is observing the female germ cells. This is correct because oogenesis is characterized by unequal cytokinesis, leading to the formation of one large ovum and small polar bodies.

The meiotic division has not completed in the correct manner. This is incorrect. The production of unequal-sized cells is the normal, correct, and evolutionarily advantageous way for oogenesis to occur. It ensures the future zygote has enough resources from the egg to begin development.

These all cells will be genetically identical. This is incorrect. Meiosis, through processes like crossing over and independent assortment, is designed to produce genetically unique cells, which is the basis of sexual reproduction.

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121. (4)

The parent organism's cells have 6n chromosomes (hexaploid).

Gametes are formed through meiosis, which is a reductional division. This means the chromosome number is halved.

Therefore, the number of chromosomes in the gamete will be: (6n) / 2 = 3n.

A zygote is formed by the fusion of two gametes.

Each gamete has 3n chromosomes.

Therefore, the number of chromosomes in the zygote will be the sum of the chromosomes from the two gametes: 3n + 3n = 6n.

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122. (2)

The question describes a chromosome that is composed of "a pair of chromatids attached to a common centromere" (i.e., a replicated chromosome) moving towards one of the poles. This event, where entire replicated chromosomes (not individual chromatids) are segregated, is the defining characteristic of Anaphase I of Meiosis. During any anaphase movement, the centromere leads the way towards the pole. The arms of the chromosome trail behind, away from the pole and towards the equatorial plate. The force that pulls chromosomes towards the poles during anaphase is generated by the disassembly and shortening of the kinetochore microtubules that are attached to the chromosome's centromere. The "second nuclear division" is Meiosis II. The event described (separation of homologous chromosomes) occurs during the first nuclear division (Meiosis I). In Meiosis II, sister chromatids separate. The "separation" of the pair of chromatids (i.e., the splitting of the centromere) occurs during Anaphase II. The stage immediately following Anaphase I is Telophase I. Therefore, this event does not happen in the "very next stage."

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123. (1)

A. Proteins are synthesised in preparation for mitosis. This occurs during the G_2 phase of Interphase. The cell is making final preparations, including synthesizing proteins like tubulin for the mitotic spindle. This is the first step in the sequence.

B. Centrosomes move towards opposite poles. This is a key event of Prophase, the first stage of mitosis (M phase). The duplicated centrosomes begin to migrate to establish the poles of the cell. This follows the G_2 phase.

D. Chromosomes align on the equatorial plate. This is the defining characteristic of Metaphase, the second stage of mitosis. All the condensed chromosomes line up in the middle of the cell. This follows Prophase.

E. Daughter chromosomes are formed. This event defines the start of Anaphase, the third stage of mitosis. The centromeres holding the sister chromatids together split, and the separated chromatids are now considered individual daughter chromosomes. This follows Metaphase.

C. Nucleolus, golgi complexes and ER reappear. This reformation of organelles and the nuclear envelope occurs during Telophase, the final stage of mitosis, as two new daughter nuclei are formed. This follows Anaphase.

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124. (1)

During zygotene, chromosomes start pairing together and this process of association is called synapsis. Such paired chromosomes are called homologous chromosomes. Electron micrographs of this stage, indicate that chromosome synapsis is accompanied by the formation of complex structure called synaptonemal complex. The complex formed by a pair of synapsed homologous chromosomes is called a bivalent or a tetrad. However, these are more clearly visible at the next stage. The first two stages of prophase I are relatively short-lived compared to the next stage that is pachytene. During this stage, the four chromatids of each bivalent chromosomes becomes distinct and clearly appear as tetrads. This stage is characterised by the appearance of recombination nodules, the sites at which crossing over occurs between non-sister chromatids of the homologous chromosomes. Crossing over is the exchange of genetic material between two homologous chromosomes. Crossing over is also an enzyme-mediated process and the enzyme involved is called recombinase. Crossing over leads to recombination of genetic material on the two chromosomes. Recombination between homologous chromosomes is completed by the end of pachytene, leaving the chromosomes linked at the sites of crossing over. The beginning of diplotene is recognised by the dissolution of the synaptonemal complex and the tendency of the recombined homologous chromosomes of the bivalents to separate from each other except at the sites of crossovers. These X-shaped structures, are called chiasmata.

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125. (3)

The S phase (Synthesis phase) of the cell cycle is when DNA replication occurs. Let's break down what happens to each component:

Before S phase (in G_1): Each chromosome consists of a single, long DNA molecule. It has one chromatid.

During S phase: The cell synthesizes an identical copy of each DNA molecule.

After S phase (in G_2): Each chromosome now consists of two identical sister chromatids joined together at a common centromere.

Therefore:

The amount of DNA in the cell has doubled.

The number of chromatids has doubled.

The cell still has only one nucleus. The nucleus divides during the M phase (karyokinesis). The number of cells only doubles after the entire M phase is complete (specifically, after cytokinesis). S or synthesis phase marks the period during which DNA synthesis or replication takes place.

During this time the amount of DNA per cell doubles. However, there is no increase in the chromosome number

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126. (4)

Anaphase I: The homologous chromosomes separate, while sister chromatids remain associated at their centromeres. Each pole receives half the chromosome number of the parent cell.

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127. (1)

The process required for repair, replacement, and general growth of somatic (body) cells is Mitosis.

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128. (2)

Anaphase I: This stage of meiosis involves the separation of homologous chromosomes. The chromosomes that move to each pole represent a complete haploid set for the organism. Determining the Haploid Number (n): The fact that 12 chromosomes were moving towards one pole tells you that the haploid number (n) for this organism is 12. The total diploid number (2n) for the parent cell was 24. A gamete is the final product of meiosis and is always a haploid cell. After Meiosis I, two cells are formed, each with 12 chromosomes. Meiosis II then occurs, where the sister chromatids of these 12 chromosomes separate. This division does not change the chromosome number.

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129. (3)

DNA synthesis: This is the defining event of the S phase of interphase. In animal cells, (II) Centriole duplication occurs in the cytoplasm simultaneously during the S phase.

Chromosomal condensation: This is the primary event of Prophase. At the same time, (I) Centriole movement to opposite poles begins, establishing the framework for the mitotic spindle.

Nuclear membrane reformation: This is a key event of Telophase, where two new daughter nuclei are formed. Simultaneously, (IV) Golgi and ER reappear as they had disintegrated during prophase. Cell furrow formation: This is the process of Cytokinesis in animal cells. As the cytoplasm divides, there is a (III) Distribution of mitochondria in daughter cells and other organelles.

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130. (2)

The stage in the question is telophase. Chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements. Nuclear envelope develops around the chromosome clusters at each pole forming two daughter nuclei. Nucleolus, golgi complex and ER reform.

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131. (3)

A typical eukaryotic cell cycle is illustrated by human cells in culture. These cells divide once in approximately every 24 hours.

However, this duration of cell cycle can vary from organism to organism and also from cell type to cell type. The term "static" means unchanging. As explained above, the cell cycle duration is one of the most variable parameters in biology, depending on the species, cell type, and developmental stage.

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132. (3)

The "subsequent stage" to Metaphase is Anaphase. In Anaphase, the centromere splits. The sister chromatid whose kinetochore is attached will be pulled towards its respective pole. The sister chromatid whose kinetochore is not attached has no connection to the spindle, so it will not be pulled anywhere. It will lag behind, likely near the center of the cell. Option (1) is incorrect because only one chromatid has pulling force. Option (2) is incorrect because the signal to split the centromere affects all chromosomes simultaneously. Option (4) is incorrect because the G_0 phase is an exit from G_1, not M-phase.

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133. (3)

In Metaphase of Mitosis, individual chromosomes (each with two chromatids) align at the metaphase plate. In the subsequent Anaphase, the centromeres split, and sister chromatids separate. This maintains the chromosome number in the daughter cells.

In Metaphase I of Meiosis, homologous chromosome pairs (bivalents) align at the metaphase plate. In the subsequent Anaphase I, the homologous chromosomes separate, but the centromeres do not split, and sister chromatids remain together. This act of separating homologous pairs is what halves the chromosome number, leading to haploid cells. This is the defining feature of the reductional division.

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134. (1)

Bivalent: A bivalent is the structure formed by a pair of synapsed homologous chromosomes during Prophase I of meiosis. Since each chromosome consists of two chromatids, the bivalent contains a total of four chromatids and is therefore also called a tetrad.

Syncytium: A syncytium is a large cell with multiple nuclei. This multinucleate condition arises when nuclear division (karyokinesis) occurs repeatedly without being followed by cytoplasmic division (cytokinesis).

Chiasmata: These are the points of contact between homologous chromosomes that are visible during the diplotene stage of Prophase I. They are the sites where crossing over occurred and have characteristic X-shaped structures.

Centromere: The centromere is the specialized region of a chromosome that holds the sister chromatids together.

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135. (4)

Leptotene: Chromosomes start to condense.

Zygotene: Homologous chromosomes pair up (synapsis) to form bivalents. At this stage, they are paired but have not yet crossed over.

Pachytene: Crossing over (the exchange of genetic material) occurs between the paired homologous chromosomes.

Diplotene: The paired chromosomes begin to separate but are held together at the points of crossing over (chiasmata).

Diakinesis: Terminalization of chiasmata occurs.

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136. (4)

Individuals with blood group O are considered "universal donors" because their red blood cells lack both A and B antigens. This means their blood won't be recognized as foreign by the recipient's immune system, regardless of the recipient's blood type.

The plasma of an individual with blood group O contains both anti-A and anti-B antibodies.

The ability of blood group O to be a universal donor is directly due to the absence of antigens (A and B) on their red blood cells.

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137. (2)

The primary function of the thin layer of pleural fluid is to act as a lubricant. This fluid is found in the pleural cavity, the space between the two pleural membranes: the visceral pleura (which covers the lungs) and the parietal pleura (which lines the inner surface of the thoracic wall).

- During breathing, the lungs expand and contract.
- The pleural fluid allows the visceral and parietal pleurae to slide smoothly over each other with minimal friction.
- This smooth gliding motion is essential to prevent injury and pain as the lungs move against the chest wall during the continuous process of respiration.

While the pleural fluid contributes to the negative intrapleural pressure that helps keep the lungs inflated, its most direct and primary function is lubrication.

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138. (3)

The Sinoatrial (SA) node generates the highest number of action potentials (70-75 per minute), making it the natural pacemaker of the heart.

The stroke volume is indeed the volume of blood (approximately 70 ml) that each ventricle pumps out during a single cardiac cycle.

The heart is autoexcitable, meaning it can generate its own electrical impulses without external neural input. However, its functions can be moderated by the nervous system (e.g., sympathetic and parasympathetic nerves) and hormones (e.g., adrenaline) to adjust the heart rate and force of contraction as needed.

The systemic heart includes the left ventricle and the left atrium. The left ventricle pumps oxygenated blood into the aorta for systemic circulation, and the right atrium receives deoxygenated blood from the systemic circulation.

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139. (3)

- Atmospheric air is primarily composed of nitrogen and oxygen. The inspired air has a high
  pO_2 (approx. 159 mmHg) and a very low pCO_2 (approx. 0.3 mmHg).
- Alveolar air has a pO_2 of about 104 mmHg and a pCO_2 of about 40 mmHg. This change in partial pressures is due to the continuous gas exchange occurring in the lungs.
- As fresh air enters the alveoli, oxygen diffuses across the respiratory membrane into the pulmonary capillaries to oxygenate the blood. This ongoing diffusion lowers the pO_2 in the alveoli.
- Simultaneously, deoxygenated blood arriving at the lungs has a high pCO_2. Carbon dioxide diffuses from the blood into the alveoli to be exhaled. This continuous diffusion increases the pCO_2 in the alveolar air.
- This constant exchange ensures the alveolar air is a mixture of newly inspired air and residual air, with partial pressures that are a result of the ongoing diffusion of gases between the alveoli and the blood.

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140. (3)

The QRS complex on an electrocardiogram (ECG) represents the depolarization of the ventricles, which is the electrical event that leads to ventricular contraction (systole). The rate at which these QRS complexes occur per minute is the ventricular rate, which is essentially the heart rate. In a healthy individual, the heart rate is equal to the pulse rate (the rhythmic expansion of arteries felt in the periphery). Therefore, by counting the QRS complexes over a period of time (e.g., on a 6-second strip and multiplying by 10), one can accurately determine the pulse rate.

The opening between the right atrium and the right ventricle is guarded by a valve formed of three muscular flaps or cusps, the tricuspid valve, whereas a bicuspid or mitral valve guards the opening between the left atrium and the left ventricle.

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141. (3)

The rate of diffusion is governed by Fick's law, which states that the rate is proportional to the surface area and the partial pressure gradient, and inversely proportional to the thickness of the membrane. The solubility of the gas is also a critical factor (as seen in CO_2 diffusing faster than O_2 despite a smaller pressure gradient).

Thickness is a key factor; a thicker membrane (as in fibrosis) reduces diffusion.

The partial pressure gradient is the primary driving force for diffusion.

Solubility is crucial; CO_2 is 20-25 times more soluble than O_2, which greatly enhances its diffusion rate.

Carbonic anhydrase is an enzyme vital for the transport of CO_2 as bicarbonate within RBCs and its subsequent release in the lungs. However, it does not directly affect the physical process of simple diffusion of gaseous CO_2 or O_2 across the alveolar membrane itself. It influences the chemical reactions that determine the partial pressure of CO_2, but it's not a direct physical factor of diffusion.

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142. (3)

- The diagram shows two cross-sections of blood vessels, A and B. Vessel A has a thicker, more muscular wall and a smaller lumen, which is characteristic of an artery. Vessel B has a thinner wall and a larger lumen, characteristic of a vein.
- The vessel connected to the left ventricle is the aorta, which carries blood for the high-pressure systemic circuit. The vessel connected to the right ventricle is the pulmonary artery, which carries blood for the low-pressure pulmonary circuit.
- The vessel entering the left atrium is the pulmonary vein, which carries oxygenated blood from the lungs. The vessel entering the right atrium is the vena cava, which carries deoxygenated blood from the body.
- The systemic artery (A) has a thick, muscular, and elastic wall to withstand the high pressure of blood being pumped from the left ventricle. The systemic vein (B) has a much thinner wall because the pressure of blood is significantly lower.
- The pulmonary artery (a type of A vessel) carries deoxygenated blood to the lungs, and the pulmonary vein (a type of B vessel) carries oxygenated blood from the lungs.

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143. (2)

- The right lung is larger and is divided into three lobes: the superior, middle, and inferior lobes. These lobes are separated by two fissures: the oblique fissure and the horizontal fissure.
- The left lung is slightly smaller to accommodate the heart and has only two lobes: the superior and inferior lobes. These are separated by a single oblique fissure.
- The composition of the blood-air barrier, also known as the respiratory membrane is the site of gas exchange in the lungs.
- The barrier is extremely thin to allow for rapid diffusion of oxygen into the blood and carbon dioxide out of the blood.
- It is a composite structure formed by three key components:

1. The simple squamous epithelium (Type I pneumocytes) of the alveolar wall.
2. The endothelium of the pulmonary capillary.
3. The fused basement membranes of the alveolar epithelium and the capillary endothelium.

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144. (2)

List-I	List-II
Blood	Most commonly used body fluid for the transport of substances
Lymph	Helps in absorption of fats
Coelenterates	Lacks a circulatory system
Plasma	55% of blood

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145. (3)

The pneumotaxic centre, located in the pons, acts as a "switch-off" point for inspiration. When the pneumotaxic centre sends strong signals to the respiratory rhythm centre, it shortens the duration of inspiration. By cutting inspiration short, it also shortens the entire respiratory cycle. A shorter cycle means more cycles can occur per minute, thus the respiratory rate increases. Therefore, a strong signal reduces the duration of inspiration and increases the respiratory rate, leading to shallower but faster breathing.

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146. (2)

Erythroblastosis fetalis is most severely caused by Rh incompatibility between a Rh-negative mother and a Rh-positive fetus. During the delivery of her first Rh-positive child, fetal red blood cells can enter the mother's circulation, causing her immune system to produce anti-Rh antibodies. These antibodies (specifically IgG) can cross the placenta in subsequent pregnancies. If the second fetus is also Rh-positive, these maternal antibodies will attack and destroy the fetus's red blood cells, leading to severe anaemia and jaundice.

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147. (1)

List-I	List-II
Earthworm	Moist Cuticle
Aquatic Arthropod	Gills
Terrestrial Reptile	Lungs
Cockroach	Network of tracheal tubes

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148. (2)

- Blood group B means the presence of B antigens (agglutinogens) on the red blood cells. The "positive" part of B-positive means the presence of the Rh antigen as well.
- A person with blood group B naturally has anti-A antibodies (agglutinins) in their serum to fight against A antigens.
- A universal recipient has no antibodies in their plasma, which is the case for someone with blood group AB. However, a person with blood group B has anti-A antibodies.
- The B antigens on the red blood cells of the donor (B-positive) would be attacked by the anti-B antibodies in the recipient's (A-positive) serum, leading to a severe haemolytic reaction.

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149. (2)

Under normal physiological conditions, the partial pressure of oxygen at the tissue level is about 40 mm Hg, which matches Point B in the graph.

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150. (2)

The most visually obvious and functionally significant difference is the presence of red blood cells (erythrocytes) in blood, which contain haemoglobin and are responsible for its red colour and oxygen-carrying capacity. Lymph, which is derived from interstitial fluid, does not contain red blood cells and is therefore a clear or yellowish fluid.

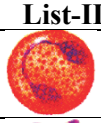
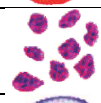
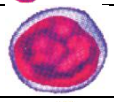
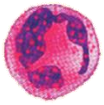
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151. (1)

Emphysema is a chronic disorder in which alveolar walls are damaged due to which respiratory surface is decreased. One of the major causes of this is cigarette smoking.

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152. (3)

List-I	List-II
Eosinophil	
Thrombocytes	
T lymphocyte	
Neutrophil	

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153. (2)

Normal, or passive, expiration is a largely passive process. It occurs when the diaphragm and external intercostal muscles relax, causing the thoracic cavity to decrease in volume and the lungs to recoil, pushing air out.

However, a forceful expiration, such as blowing out a candle or coughing, is an active process that requires the contraction of specific muscles to expel a greater volume of air.

- Contraction of the diaphragm: The diaphragm is the primary muscle of inspiration. During quiet expiration, it simply relaxes and moves upward. During forceful expiration, it relaxes even more as other muscles push it up.
- Contraction of external intercostal muscles: The external intercostal muscles are also muscles of inspiration. They contract to raise the rib cage and expand the thoracic cavity. During forceful expiration, they relax.
- Contraction of abdominal muscles: The abdominal muscles (like the rectus abdominis and internal/external obliques) contract forcefully. This pushes the abdominal organs upward against the diaphragm.
- An increase in intra-abdominal pressure: The contraction of the abdominal muscles (C) directly leads to an increase in intra-abdominal pressure. This pressure is transmitted to the diaphragm, pushing it further into the thoracic cavity and helping to compress the lungs to expel air.
- Contraction of internal intercostal muscles: The internal intercostal muscles are the primary muscles of forceful expiration. They contract to pull the ribs down and inward, further decreasing the volume of the thoracic cavity and increasing intrathoracic pressure to force air out.

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154. (3)

- The complete separation of the pulmonary (to the lungs) and systemic (to the body) circuits. This separation guarantees that the blood delivered to the body's tissues is maximally saturated with oxygen, which is essential for organisms with high metabolic demands, like birds and mammals.
- A key advantage of a closed circulatory system is in this the blood is contained within vessels, the pressure can be kept high, and flow can be shunted towards or away from specific tissues (e.g., to muscles during exercise and away from the digestive system) far more effectively than in an open system where blood (haemolymph) flows slowly and bathes all tissues in a body cavity.

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155. (2)

The correct pathway for inhaled air is: External Nostrils → Nasal Passage → Nasal Chamber → Pharynx (Nasopharynx, Oropharynx, Laryngopharynx) → Larynx → Trachea → Primary Bronchi → Secondary Bronchi → Tertiary Bronchi → Bronchioles → Terminal Bronchioles → Respiratory Bronchioles → Alveolar Ducts → Alveoli.

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156. (3)

- A healthy human has an average of 5 to 5.5 million RBCs per cubic millimeter of blood, while the WBC count is only about 6,000 to 8,000 per cubic millimeter. This makes RBCs far more numerous.
- Mature mammalian RBCs are biconcave and enucleated (lacking a nucleus). This structural feature allows them to carry more oxygen. In contrast, all types of WBCs are nucleated and also contain other organelles.
- The red color of RBCs is due to the presence of the iron-containing pigment haemoglobin, which is essential for oxygen and carbon dioxide transport. WBCs lack this pigment, making them colorless, hence the name "white blood cells."
- RBCs have a lifespan of approximately 120 days and are then destroyed primarily in the spleen. WBCs have a much shorter lifespan, ranging from a few hours to a few days, depending on their type.

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157. (2)

Food must be directed into the esophagus, not the trachea (windpipe) via the larynx, to prevent choking and aspiration into the lungs. This is a critical protective mechanism.

The is the opening into the larynx. During swallowing, the larynx moves up, and the (a leaf-shaped piece of cartilage) flips down to cover the .

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158. (2)

A standard Electrocardiogram (ECG) measures the electrical activity of the heart during a cardiac cycle and shows three main peaks or waves.

- The P-wave represents the depolarization of the atria, which leads to their contraction.
- The QRS complex represents the depolarization of the ventricles, which initiates their contraction. The repolarization of the atria also occurs during this time but is masked by the larger QRS complex.
- The T-wave represents the repolarization of the ventricles, when they return to a resting state.

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159. (1)

TV ≈ 500 mL; IRV ≈ 3000 mL; ERV ≈ 1100 mL; RV ≈ 1200 mL.

Inspiratory Capacity (IC) = TV + IRV = 500 + 3000 = 3500 mL.

Functional Residual Capacity (FRC) = ERV + RV = 1100 + 1200 = 2300 mL.

Vital Capacity (VC) = TV + IRV + ERV = 500 + 3000 + 1100 = 4600 mL.

Total Lung Capacity (TLC) = VC + RV = 4600 + 1200 = 5800 mL.

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160. (3)

- Hypertension is also known as "high blood pressure," but it is a chronic condition, not a transient one. It's defined as a consistently elevated blood pressure over a long period.
- The hepatic portal vein carries nutrient-rich blood from the intestine to the liver, not the other way around. This system allows the liver to process and detoxify substances absorbed from the digestive tract before they enter the general circulation.
- The coronary circulation is a dedicated system of arteries and veins that supplies oxygen and nutrients to the heart muscle (myocardium) itself.
- The pericardium is the double-walled sac that encloses the heart. It is a fibrous protective covering, not muscular. The heart muscle itself is the myocardium.

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161. (4)

List-I	List-II
Spirometer	Cannot measure residual volume
Inspiration	Diaphragm flattens
Pharynx	Common route for air and food
Secondary bronchioles	Have cartilaginous rings

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162. (2)

The delay at the AV node is essential. It ensures that the atria have finished contracting and have completely filled the ventricles with blood before the ventricles are stimulated to contract. This sequence maximizes the efficiency of each heartbeat.

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163. (2)

Slow, deep breathing is the most efficient because it minimizes the percentage of each breath that is "wasted" in the anatomic dead space (the conducting airways where no gas exchange occurs). A larger tidal volume ensures that more fresh air reaches the alveoli with each breath, maximizing the amount of air available for gas exchange relative to the effort of breathing.

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164. (1)

- The complexity of circulatory systems does indeed vary greatly across the animal kingdom. Simple organisms like sponges rely on water flowing through their bodies, while more complex animals have evolved specialized systems with pumps (hearts) and vessels to move blood.
- Every living cell within an organism must be supplied with nutrients and oxygen and have its metabolic byproducts cleared away to maintain homeostasis and survive. This fundamental requirement is what drives the evolution of circulatory systems in the first place.

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165. (1)

Increased metabolism produces more CO_2 and lactic acid, which lowers blood pH.

The body increases the rate and depth of breathing to take in more O_2 and expel the excess CO_2.

Higher temperature, increased CO_2, and lower pH in the muscles cause the oxygen-haemoglobin curve to shift to the right, which helps release oxygen more easily to the active tissues.

Tissues consume O_2, so its partial pressure decreases.

Blood vessels to muscles vasodilate (widen) to increase blood flow.

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166. (2)

The sympathetic nervous system (SNS) and adrenaline (epinephrine) are both part of the body's integrated stress response and have similar, reinforcing (synergistic) effects on the heart.

- Sympathetic nerves provide a rapid, direct stimulation to the SA node (increasing the rate of firing) and the ventricular myocardium (increasing contractility).
- Adrenaline, released into the bloodstream, provides a slightly slower but more widespread and sustained hormonal stimulation that also increases both heart rate and force of contraction.
- The combined result is a powerful and lasting increase in cardiac output (Cardiac Output = Heart Rate × Stroke Volume), preparing the body for intense physical activity.

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167. (2)

The respiratory control centers in the brain are most sensitive to changes in arterial PCO_2 and the resulting changes in pH. Even a small increase in PCO_2 strongly stimulates central chemoreceptors, leading to an increase in breathing rate and depth to expel the excess CO_2. The role of PO_2 is secondary and only becomes a significant stimulus when it drops to very low levels.

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168. (2)

The heart of a highly trained athlete adapts to be more efficient.

- The cardiac muscle (myocardium) becomes stronger and the ventricular chamber larger. This allows the heart to pump a much larger volume of blood with each beat (increased stroke volume).
- Because each beat is so effective at ejecting blood, the heart does not need to beat as often at rest to maintain a normal cardiac output (approx. 5 L/min). This results in a slower resting heart rate, often below 60 bpm (bradycardia).
- A slower heart rate means the total duration of the cardiac cycle is longer. This extra time is predominantly added to the filling phase (ventricular diastole), allowing the ventricles to fill more completely, which further contributes to the high stroke volume

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169. (2)

List-I	List-II
Normal breathing rate	12-16 times/minute
Epiglottis	Thin elastic cartilaginous flap
100 ml oxygenated blood	Can deliver around 5 ml of O_2 to the tissues under normal physiological conditions.
Larynx	Sound box

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170. (1)

Correct sequence of clotting is as follows:

1. First, an injury stimulates platelets/tissues to release factors (III).
2. This initiates a series of reactions that form the enzyme complex, thrombokinase (I).
3. Thrombokinase then catalyzes the conversion of prothrombin into thrombin (IV).
4. Thrombin, in turn, acts as an enzyme to convert fibrinogen into fibrin threads (II).
5. Finally, these fibrin threads create a mesh that traps blood cells, forming the clot or coagulum (V).

Therefore, the correct sequence is C → A → D → B → E.

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171. (3)

The values $PO_2 \approx 95$ mmHg and $PCO_2 \approx 40$ mmHg are characteristic of arterial blood. Systemic tissues use O_2 and produce CO_2, creating these partial pressures. The deoxygenated blood leaving these tissues, which then travels through the pulmonary artery, reflects these same values.

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172. (3)

Blood is unique among connective tissues because its extracellular matrix is a liquid called plasma. Within this fluid matrix are the cellular components, which are collectively known as formed elements. These include:

- Erythrocytes (red blood cells)
- Leucocytes (white blood cells)
- Platelets

This composition—cells suspended in an extracellular matrix—is the defining characteristic of all connective tissues.

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173. (2)

While someCO_2 is transported dissolved in the plasma, this accounts for only about 7-10% of the total CO_2 transport. The majority of CO_2 (about 70%) is transported as bicarbonate ions (HCO_3^-), which are formed primarily inside red blood cells (RBCs) and then diffuse into the plasma.

Carbonic anhydrase is the enzyme that catalyzes the rapid conversion of CO_2 and water (H_2O) into carbonic acid (H_2CO_3), which then dissociates into HCO_3^- and hydrogen ions (H^+). This reaction is extremely fast inside RBCs due to the high concentration of the enzyme. The reaction also occurs in the plasma, but at a much slower rate because the enzyme is present in only minute quantities there.

The binding of CO_2 to haemoglobin to form carbamino-haemoglobin is primarily related to the partial pressure of CO_2. While pO_2 does have an inverse effect (Haldane effect), pCO_2 is the primary factor.

Every 100 ml of deoxygenated blood delivers approximately 4 ml of CO_2 to the alveoli.

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174. (3)

- Papillary muscles are cone-shaped muscles located in the ventricles, and the chordae tendineae are fibrous cords that connect the papillary muscles to the cusps of the atrioventricular (AV) valves (tricuspid and bicuspid valves).
- During ventricular systole (contraction), the pressure inside the ventricles rises significantly. This pressure pushes the AV valve cusps closed.
- Simultaneously, the papillary muscles contract, which tenses the chordae tendineae. This tension acts like guy wires, pulling on the valve cusps and preventing them from turning inside out and prolapsing into the atria. This is crucial for ensuring unidirectional blood flow from the atria to the ventricles and then out of the heart.
- The AV valves open passively when atrial pressure is higher than ventricular pressure. They are not pulled open by the chordae tendineae and papillary muscles.
- The right atrium and ventricle are separated by tricuspid valve having three muscular flaps or cusps unlike the mitral or bicuspid valve present between left atrium and ventricle having two muscular flaps or cusps.
- While the heart valves ensure unidirectional blood flow, their primary role is not to prevent the mixing of oxygenated and deoxygenated blood. That is a function of the heart's septum, which separates the right and left sides.

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175. (2)

- "Lub" is the first heart sound (S_1). It is produced by the closure of the atrioventricular (AV) valves (the tricuspid and bicuspid valves) at the beginning of ventricular systole (contraction).
- "Dub" is the second heart sound (S_2). It is produced by the closure of the semilunar valves (the aortic and pulmonary valves) at the end of ventricular systole.

These two sounds happen in sequence as the heart pumps blood, forming the characteristic "Lub-Dub" sound of a single heartbeat.

- The apex is the pointed, inferior end of the heart. It is directed downwards, anteriorly, and to the left, resting on the diaphragm. The base of the heart is the broader, superior end, where the major blood vessels (aorta, vena cava, pulmonary arteries, and veins) enter and exit.

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176. (3)

- Atherosclerosis is a condition where arteries narrow due to the buildup of plaque, which is composed of cholesterol, fatty substances, and calcium. This process hardens and thickens the arterial walls, restricting blood flow.
- Angina pectoris is characterized by chest pain that occurs when the heart muscle doesn't receive enough oxygen. It's a warning sign of an underlying issue like coronary artery disease, but it's not a heart attack (myocardial infarction). A heart attack is when blood flow to the heart muscle is completely blocked, leading to tissue death.
- Heart failure is a chronic condition in which the heart can't pump enough blood to meet the body's needs, but it doesn't stop working. Cardiac arrest is a sudden, abrupt loss of heart function, where the heart's electrical system malfunctions and it stops pumping blood, which is a medical emergency.
- A primary symptom of congestive heart failure is the congestion of the lungs. The term "congestive" refers to the fluid backup that occurs when the left side of the heart is too weak to pump blood effectively. This blood backs up into the lungs, causing fluid to leak into the lung tissue, leading to shortness of breath and coughing.

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177. (3)

- Gas moves passively from an area of higher pressure to an area of lower pressure. Inspiration occurs when intrapulmonary pressure is lower than atmospheric pressure, and expiration occurs when intrapulmonary pressure is higher than atmospheric pressure. This pressure gradient is the immediate driving force for air movement.
- The lungs themselves have no muscle to drive this movement. The primary respiratory muscles (diaphragm and intercostals) contract (inspiration) or relax (expiration) to change the volume of the thoracic cavity.

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178. (2)

Human platelets are anucleate cytoplasmic fragments derived from megakaryocytes, crucial for blood clotting. They are not involved in oxygen transport (RBCs do that), not pathogen defense (WBCs do that), and not multinucleated.

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179. (3)

The conducting part (from external nostrils to terminal bronchioles) is responsible for transporting the air, filtering it from dust and foreign particles, humidifying it (adding moisture), and bringing it to body temperature. The respiratory or exchange part (which consists of the alveoli and their ducts) is where the actual exchange of gases (diffusion of O_2 and CO_2) between the blood and atmospheric air occurs.

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180. (3)

- The outer pleural membrane is in close contact with the thoracic lining whereas the inner pleural membrane is in contact with the lung surface.
- The trachea and bronchi are supported by C-shaped cartilaginous rings. The open part of the "C" faces the oesophagus, which lies posterior to the trachea.
- During inspiration, the pressure inside the airways drops. These cartilaginous rings provide structural support, preventing the trachea and bronchi from collapsing due to this negative pressure.
- The incomplete nature of the rings allows the oesophagus to expand into the space behind the trachea when a bolus of food is being swallowed. This expansion is essential for deglutition (swallowing).

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